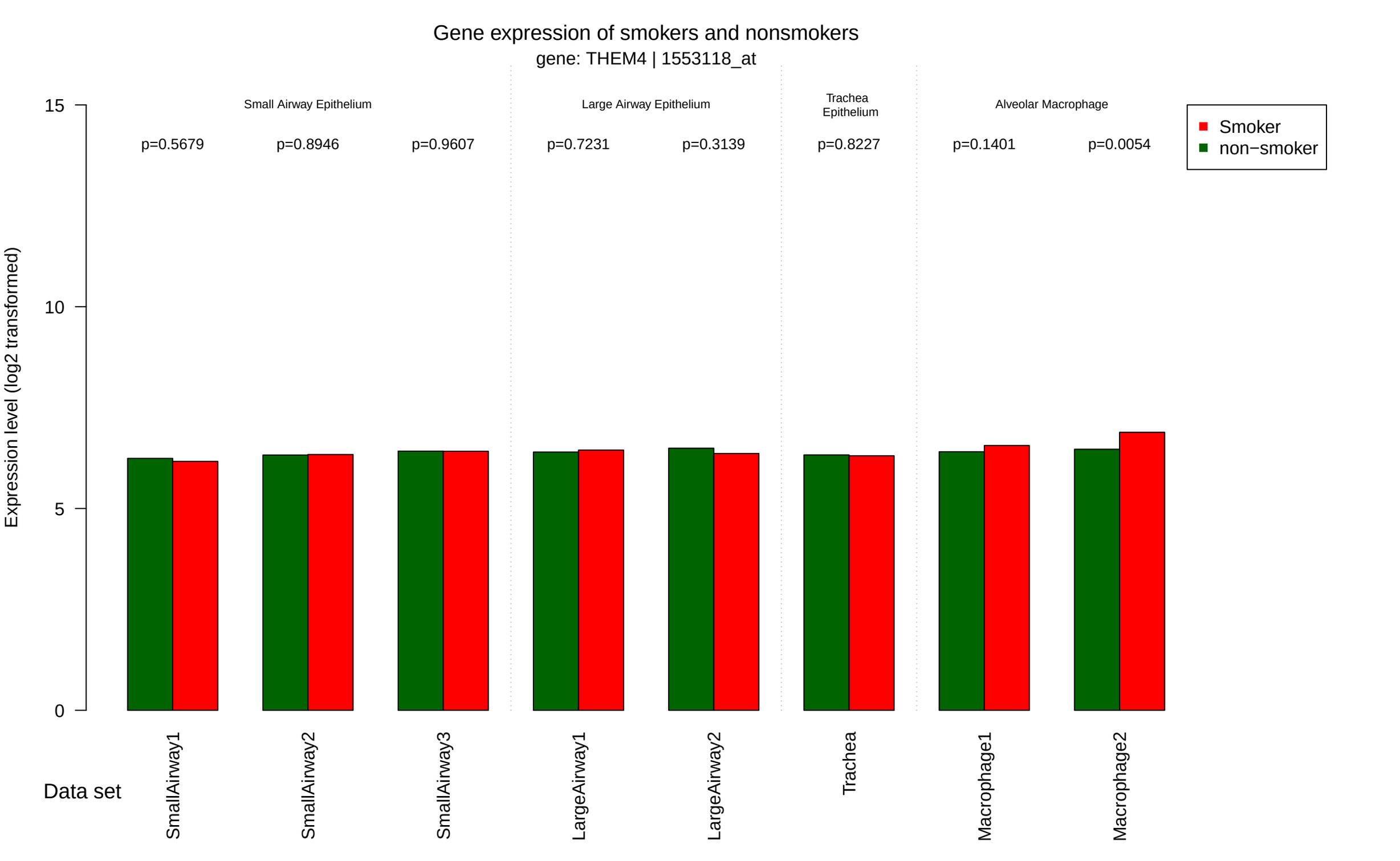
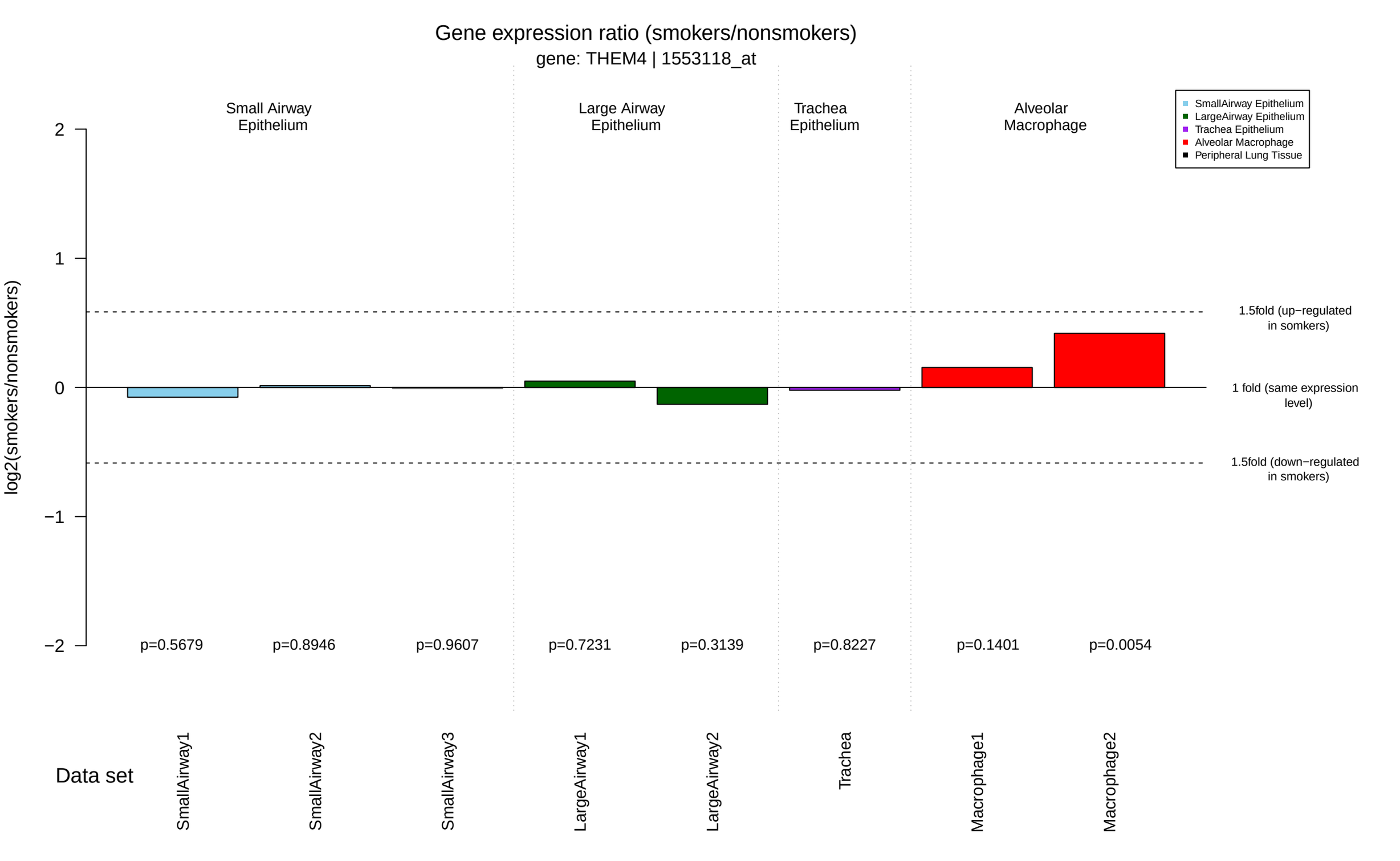
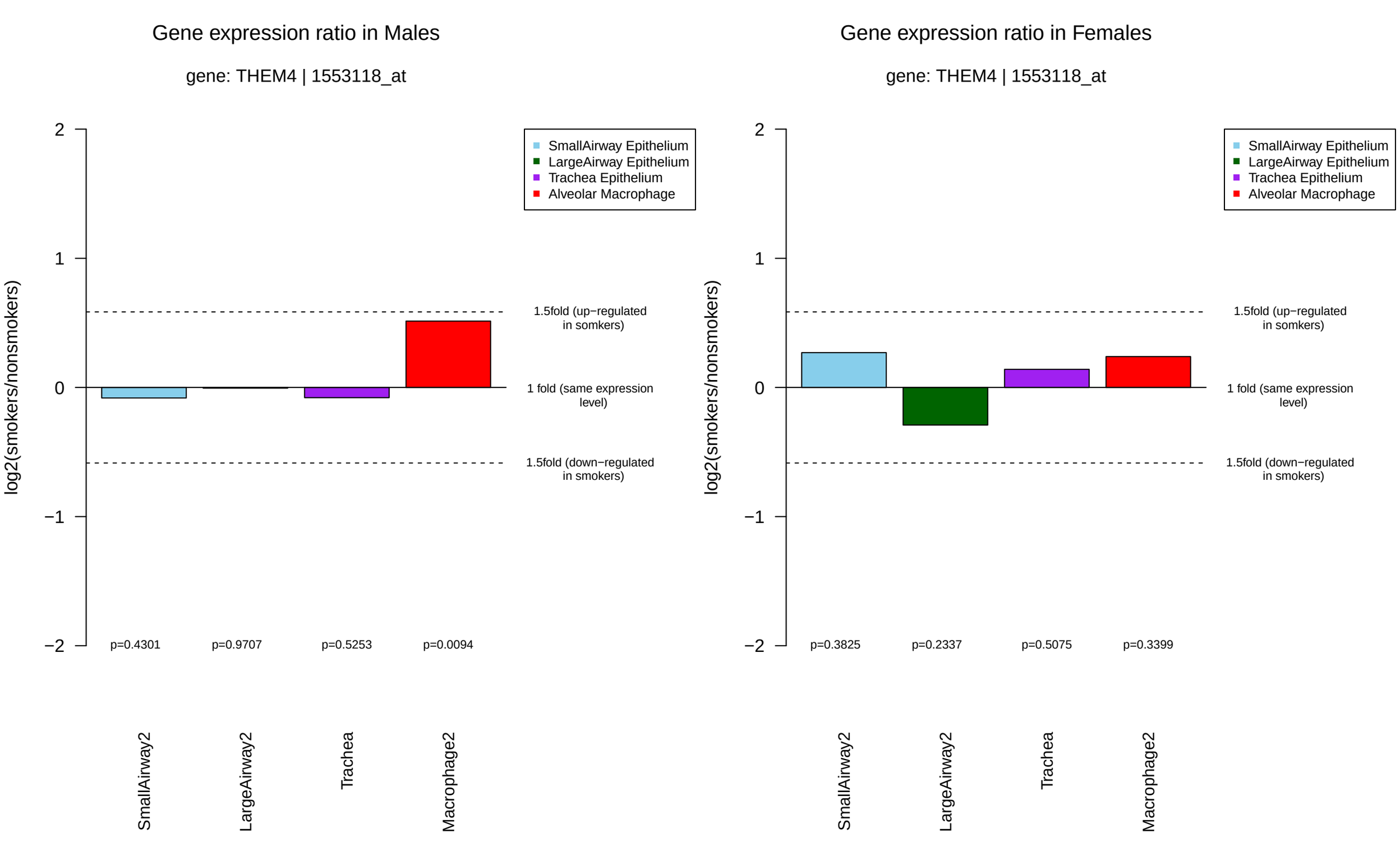
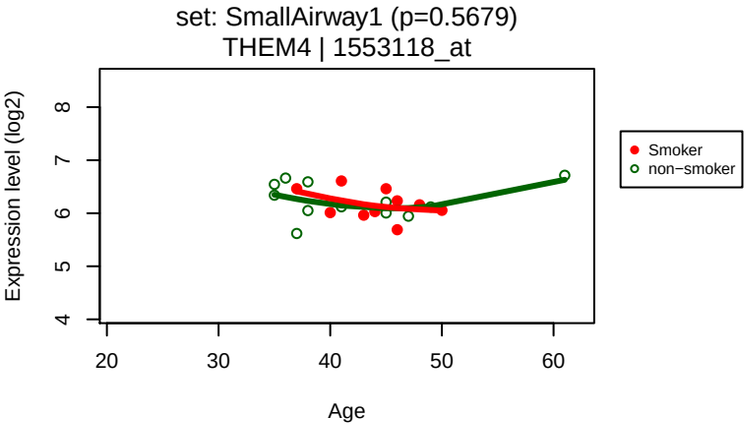
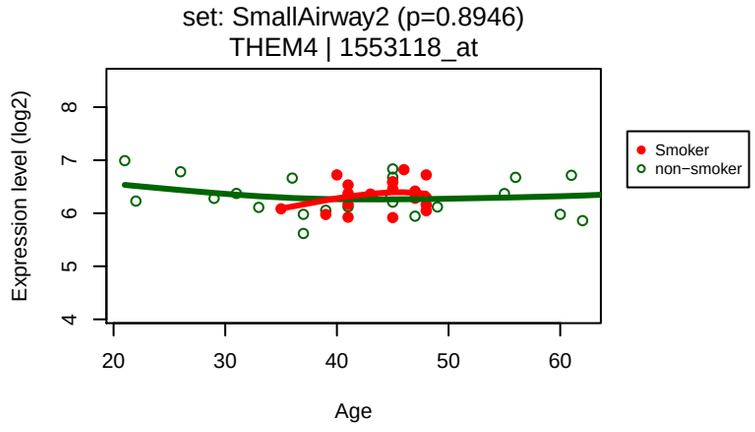
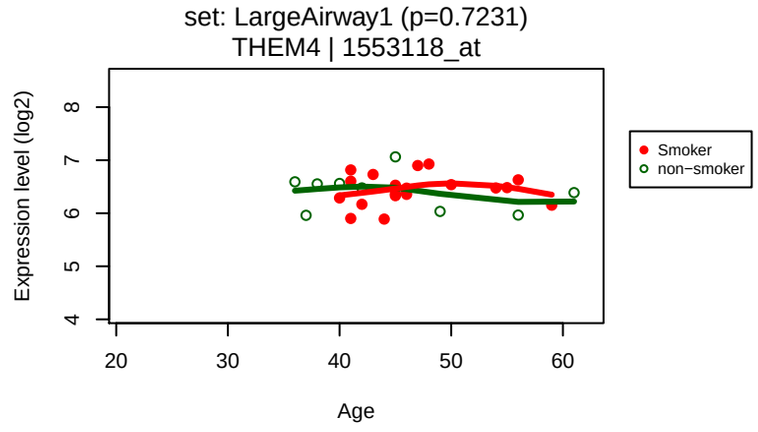

Gene expression of smokers and nonsmokers
gene: THEM4 | 1553118_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5679
p=0.8946
p=0.9607
p=0.7231
p=0.3139
p=0.8227
p=0.1401
p=0.0054
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

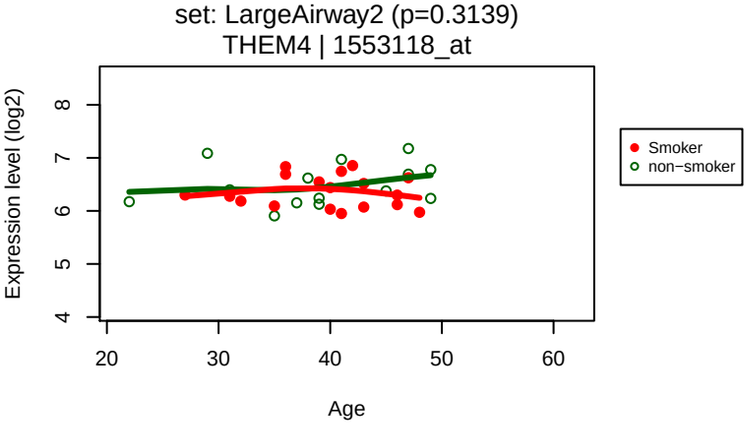

Gene expression ratio (smokers/nonsmokers)
gene: THEM4 | 1553118_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5679
p=0.8946
p=0.9607
p=0.7231
p=0.3139
p=0.8227
p=0.1401
p=0.0054
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

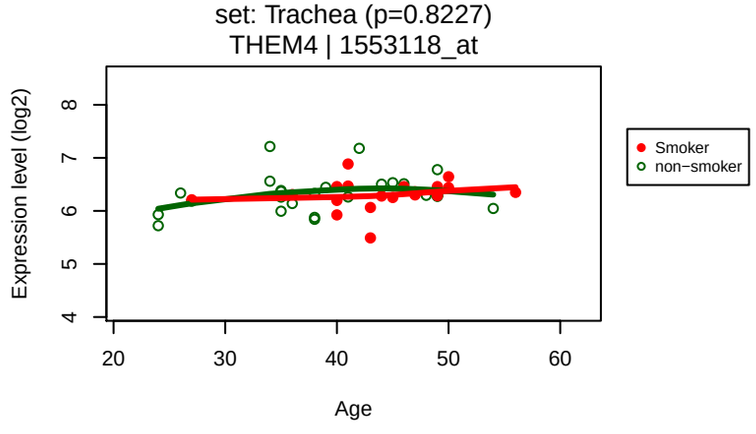

Gene expression ratio in Males
gene: THEM4 | 1553118_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4301
p=0.9707
p=0.5253
p=0.0094
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: THEM4 | 1553118_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3825
p=0.2337
p=0.5075
p=0.3399
SmallAirway2
LargeAirway2
Trachea
Macrophage2

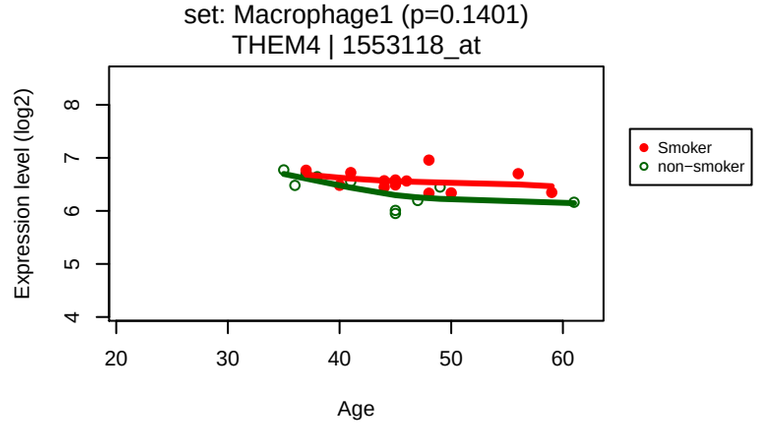

set: SmallAirway1 (p=0.5679)
THEM4 | 1553118_at
Expression level (log2)
Age
Smoker
non−smoker

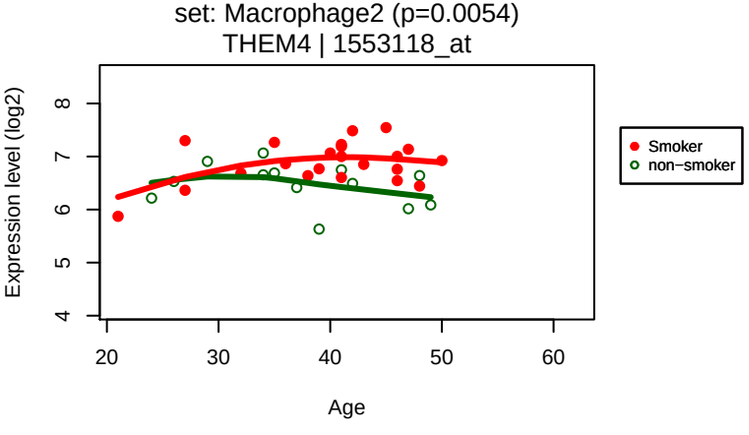

set: SmallAirway2 (p=0.8946)
THEM4 | 1553118_at
Expression level (log2)
Age
Smoker
non−smoker

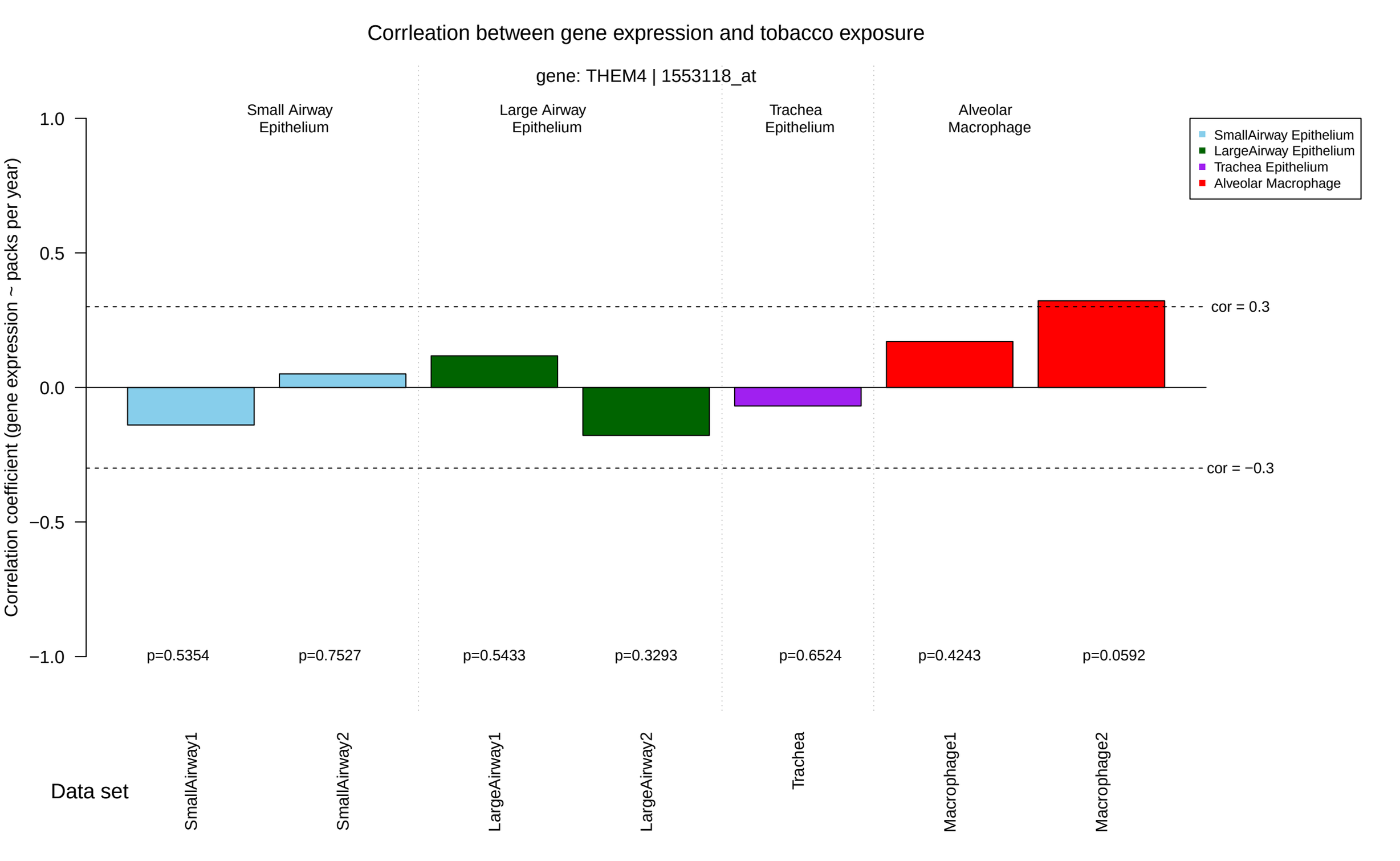

set: LargeAirway1 (p=0.7231)
THEM4 | 1553118_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3139)
THEM4 | 1553118_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8227)
THEM4 | 1553118_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1401)
THEM4 | 1553118_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0054)
THEM4 | 1553118_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: THEM4 | 1553118_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5354
p=0.7527
p=0.5433
p=0.3293
p=0.6524
p=0.4243
p=0.0592
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2